CHE 109: Engineering Chemistry – I
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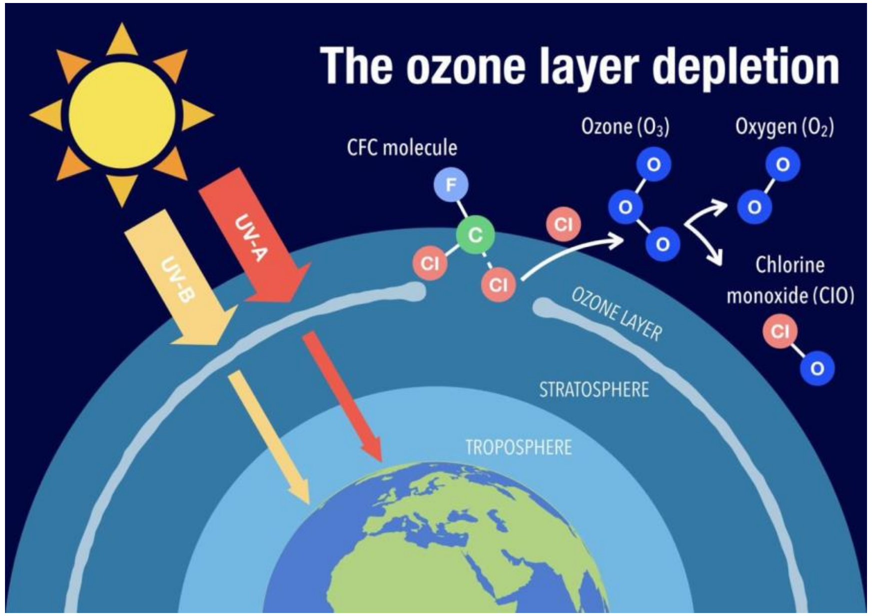
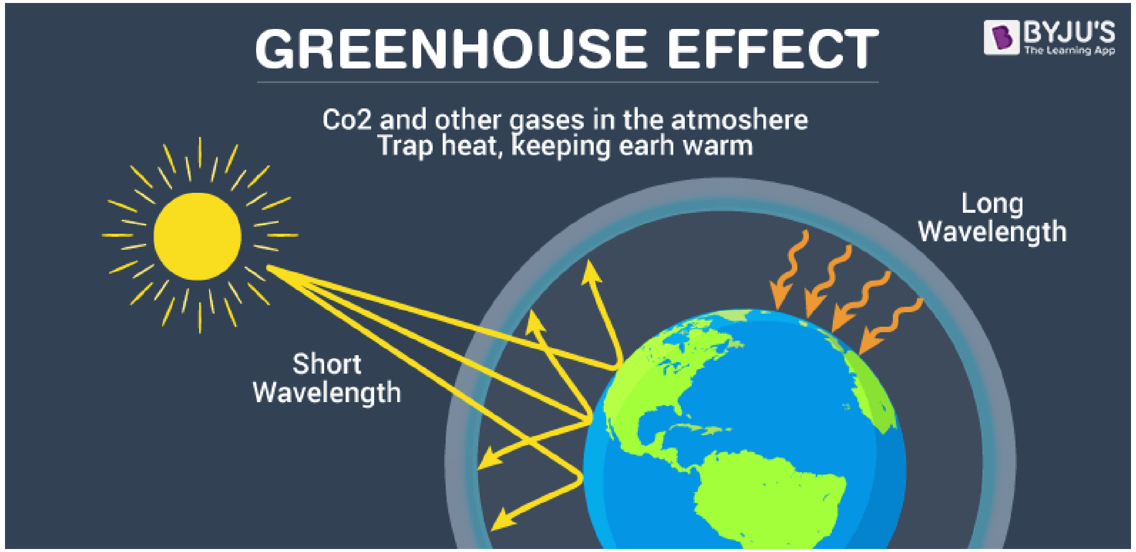
Environmental Chemistry
Chapter 8

Prof. Dr. Md. Kamal Hossain
Adjunct Faculty
East West University


The ozone layer depletion
CFC molecule
Ozone (O_3)
Oxygen (O_2)
Chlorine
monoxide (ClO)
UV-A
UV-B
OZONE LAYER
STRATOSPHERE
TROPOSPHERE


ACID RAIN
wind
H_2O
H_2SO_4
HNO_3
NO_2
SO_2
evaporation


GREENHOUSE EFFECT
BYJU'S
The Learning App
Co2 and other gases in the atmoshere
Trap heat, keeping earh warm
Long
Wavelength
Short
Wavelength

Environmental pollution:

- Environmental pollution is one of the most serious problems facing humanity and other life forms on our planet today.

- "Environmental pollution is defined as "the contamination of the physical and biological components of the earth/atmosphere system to such an extent that normal environmental processes are adversely affected."

- Environmental pollution is of different types namely air, water, soil, noise and light-weight. These cause damage to the living system. How pollution interacts with public health, environmental medicine and the environment has undergone dramatic change

Air pollution:

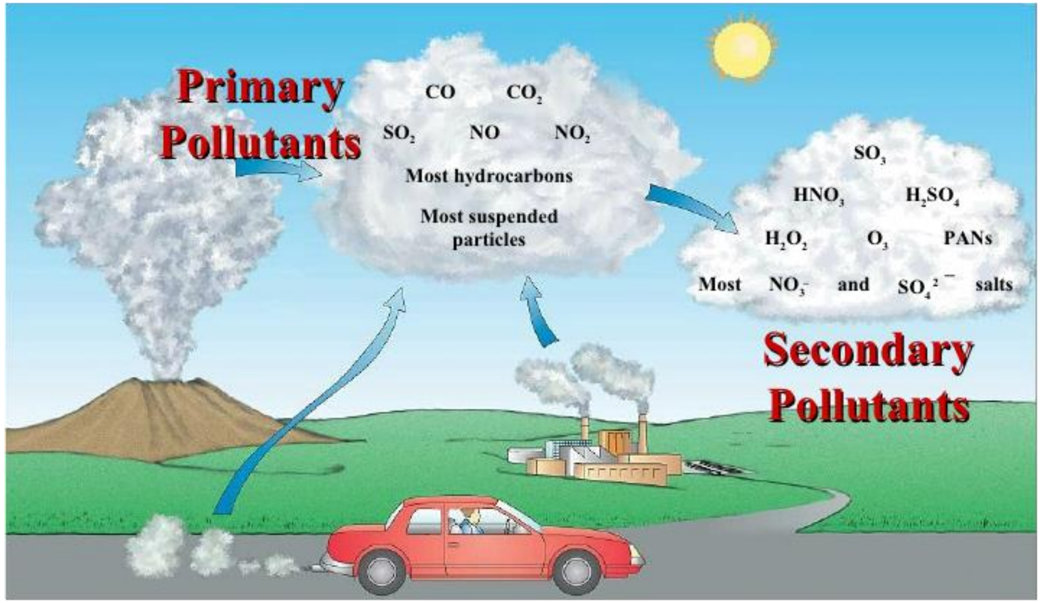
- Air pollution is a mixture of solid particles and gases in the air.

- Air pollution occurs when harmful or excessive quantities of substances are introduced into Earth's atmosphere.

- Sources of air pollution include gases, particulates, and biological molecules."

Primary and secondary air pollutants:

- A primary pollutant is an air pollutant emitted directly from a source. A secondary pollutant is not directly emitted as such, but forms when other pollutants (primary pollutants) react in the atmosphere.



- The primary pollutants are “directly” emitted from the processes such as fossil fuel consumption, volcanic eruption and factories.

- The major primary pollutants are Oxides of Sulfur, Oxides of Nitrogen, Oxides of Carbon, Particulate Matter, Methane, Ammonia, Chlorofluorocarbons, Toxic metals etc.

- Examples of Primary Pollutants: 1. Car exhaust, smokestacks (CO, SO_2, NO) 2. Particulate material (soot, ash) 3. Toxic metals (lead, mercury) 4. Volatile organic compounds (VOCs) (methane, propane, CFCs, etc.)

Secondary air pollutants:

- The secondary pollutants are not emitted directly. The secondary pollutants form when the primary pollutants react with themselves or other components of the atmosphere. Most important secondary level Air Pollutants are Ground Level Ozone, Smog and POPs (Persistent Organic Pollutants).

Causes of air pollution:

Burning of Fossil Fuels: The combustion of fossil fuels emits a large amount of sulfur dioxide. Carbon monoxide released by incomplete combustion of fossil fuels also results in air pollution.

Automobiles: The gases emitted from vehicles such as jeeps, trucks, cars, buses, etc.

Agricultural Activities: Ammonia is one of the most hazardous gases emitted during agricultural activities.

Factories and Industries: Factories and industries are the main source of carbon monoxide, organic compounds, hydrocarbons, and chemicals.

Mining Activities: In the mining process, the minerals below the earth are extracted using large pieces of equipment.

Domestic Sources: The household cleaning products and paints contain toxic chemicals that are released in the air. The smell from the newly painted walls is the smell of the chemicals present in the paints. It not only pollutes the air but also affects breathing.

Effects of Air Pollution:

Diseases: Air pollution has resulted in several respiratory disorders and heart diseases among humans. The cases of lung cancer have increased in the last few decades. Children living near polluted areas are more prone to pneumonia and asthma.

Global Warming: Due to the emission of greenhouse gases, there is an imbalance in the gaseous composition of the air. This has led to an increase in the temperature of the earth.

Acid Rain: The burning of fossil fuels releases harmful gases such as nitrogen oxides and sulfur oxides in the air. The water droplets combine with these pollutants, become acidic, and fall as acid rain which damages human, animal and plant life.

Ozone Layer Depletion: The release of chlorofluorocarbons (CFC), halons, and hydrochlorofluorocarbons in the atmosphere is the major cause of depletion of the ozone layer.

Effect on Animals: The air pollutants suspend on the water bodies and affect the aquatic life. Pollution also compels the animals to leave their habitat and shift to a new place.

Air Pollution Control:

Following are the measures one should adopt to control air pollution:

- **Avoid Using Vehicles:** People should avoid using vehicles for shorter distances. Rather they should prefer public modes of transport to travel from one place to another. This not only prevents pollution but also conserves energy.

- **Energy Conservation:** A large number of fossil fuels are burnt to generate electricity. Therefore, do not forget to switch off the electrical appliances when not in use. Thus, you can save the environment at the individual level. Use of energy-efficient devices such compact fluorescent lamp (CFLs) also controls pollution to a greater level.

- **Use of Energy efficient appliances:** Whether at the domestic level or at the industrial level, we must push for appliances that use energy efficiently, which result in complete combustion of fuel, as incomplete combustion causes air pollution.

- **Shifting industries:** Another possible solution to reduce the harmful effects of air pollution is to shift the manufacturing plants, factories and industries to remote areas with a low level of population.

Using Modern Techniques: With technology making great advancements, there are now technologies available that can help reduce the release of pollutants in the air. Air filters, scrubbers, precipitators are just a few examples.

Shifting to Natural Gasses: Instead of using and exhausting fossil fuels, shifting to greener options is a no-brainer. For example, using CNG (compressed natural gas) instead of petrol or diesel is a great option.

Water pollution:

- Water pollution is the contamination of water bodies, usually as a result of human activities. Water bodies include for example lakes, rivers, oceans, aquifers and groundwater.

- Water pollution results when contaminants are introduced into the natural environment.

Sources of Water Pollution:

Point and non-point sources:

1. When pollutants are discharged from a specific location such as a drain pipe carrying industrial effluents discharged directly into a water body it represents point source pollution

2. In contrast, non-point sources include discharge of pollutants from diffused sources or from a larger area such as runoff from agricultural fields, grazing lands, constriction site, abandoned mines and pits, etc.

Causes of Water Pollution:

The causes of water pollution vary and may be both natural and anthropogenic. However, the most common causes of water pollution are the anthropogenic ones, including:

Agrochemicals: Agrochemicals like fertilizers (containing nitrates and phosphates) and pesticides (insecticides, fungicides, herbicides etc.) washed by rain-water and surface runoff pollute water.

Storm water runoff:
Carrying various oils, petroleum products, and other contaminants from urban and rural areas (ditches). These usually forms sheens on the water surface.

Sewage: Emptying the drains and sewers in fresh-water bodies causes water pollution. The problem is severe in cities.

Mining activities: Mining activities involve crushing rocks that usually contain many trace metals and sulfides. The leftover material from mining activities may easily generate sulfuric acid in the presence of precipitation water.

Industrial Effluents: Industrial wastes containing toxic chemicals, acids, alkalis, metallic salts, phenols, cyanides, ammonia, radioactive substances, etc., are sources of water pollution. They also cause thermal (heat) pollution of water.

Burning of fossil fuels: The emitted ash particles usually contain toxic metals (such as As or Pb). Burning will also add a series of oxides including carbon dioxide to air and, respectively, water bodies.

Leaking landfills: May pollute the groundwater below the landfill with a large variety of contaminants (whatever is stored by the landfill).

Animal waste:

Contribute to the biological pollution of water streams. Think of it this way: anything that can cause air pollution or soil pollution may also affect water bodies and cause innumerable ecological and human health issues

Effects of water pollution:

- The effects of water pollution are varied. They include poisonous drinking water, poisonous food animals (due to these organisms having bioaccumulated toxins from the environment over their life spans), unbalanced river and lake ecosystems that can no longer support full biological diversity, deforestation from acid rain, and many other effects.

- These effects are, of course, specific to the various contaminants.

- Water bodies in the vicinity of urban areas are extremely polluted. This is the result of dumping garbage and toxic chemicals by industrial and commercial establishments.

- Water pollution drastically affects aquatic life. It affects their metabolism, behavior, causes illness and eventual death. Dioxin is a chemical that causes a lot of problems from reproduction to uncontrolled cell growth or cancer. This chemical is bioaccumulated in fish, chicken and meat. Chemicals such as this travel up the food chain before entering the human body.

- The effect of water pollution can have a huge impact on the **food chain.** It disrupts the food chain. Cadmium and lead are some toxic substances, these pollutants upon entering the food chain through animals (fish when consumed by animals, humans) can continue to disrupt at higher levels.

- Humans are affected by pollution and can contract diseases such as hepatitis through faucal matter in water sources.

- The ecosystem can be critically affected, modified and destructured because of water pollution.

Control of Water Pollution:

- Water pollution, to a larger extent, can be controlled by a variety of methods. Rather than releasing sewage waste into water bodies, it is better to treat them before discharge.

- Practicing this can reduce the initial toxicity and the remaining substances can be degraded and rendered harmless by the water body itself.

- If the secondary treatment of water has been carried out, then this can be reused in sanitary systems and agricultural fields.

- Some chemical methods that help in the control of water pollution are precipitation, the ion exchange process, reverse, and coagulation. As an individual, reusing, reducing, and recycling wherever possible will advance a long way in overcoming the effects of water pollution.

Acid rain refers to any form of precipitation (rain, snow, fog, or dust) that has a higher level of acidity than normal, due to the presence of pollutants like sulfur dioxide (SO_2) and nitrogen oxides (NO_x) in the atmosphere. These pollutants are primarily released by the burning of fossil fuels, industrial processes, and vehicle emissions. When these pollutants combine with water vapor in the atmosphere, they form sulfuric acid (H_2SO_4) and nitric acid (HNO_3), which then fall to the ground as acid rain.

Effects of Acid Rain:

1.Environmental Damage:

1. **Water bodies**: Acid rain can lower the pH of lakes, rivers, and streams, harming aquatic life, including fish and other organisms that are sensitive to changes in pH.
2. **Soil**: It can leach important minerals from the soil, making it harder for plants to grow and affecting forests and agricultural crops.
3. **Vegetation**: Acid rain can directly damage plant leaves and roots, impairing their ability to photosynthesize and grow.

2.Building and Infrastructure Damage:

1. Acid rain can erode buildings, monuments, and statues, particularly those made of limestone, marble, or other calcium-based stones, by reacting with calcium carbonate and weakening the structure.

3.Human Health:

1. Although acid rain itself doesn't pose a direct health risk to humans, the pollutants that cause it (e.g., sulfur dioxide and nitrogen oxides) can contribute to respiratory problems, including asthma and bronchitis.

Efforts to Combat Acid Rain:

•**Clean Air Acts**: In many countries, including the United States, regulations have been put in place to limit emissions of sulfur dioxide and nitrogen oxides. For example, the U.S. Clean Air Act Amendments of 1990 established programs to reduce these pollutants.

•**Renewable Energy and Cleaner Technologies**: Transitioning to cleaner energy sources (such as wind, solar, and natural gas) helps reduce the amount of sulfur dioxide and nitrogen oxides released into the atmosphere.

In summary, acid rain is a serious environmental problem, but it has been mitigated in some areas through stronger regulations and cleaner technologies.